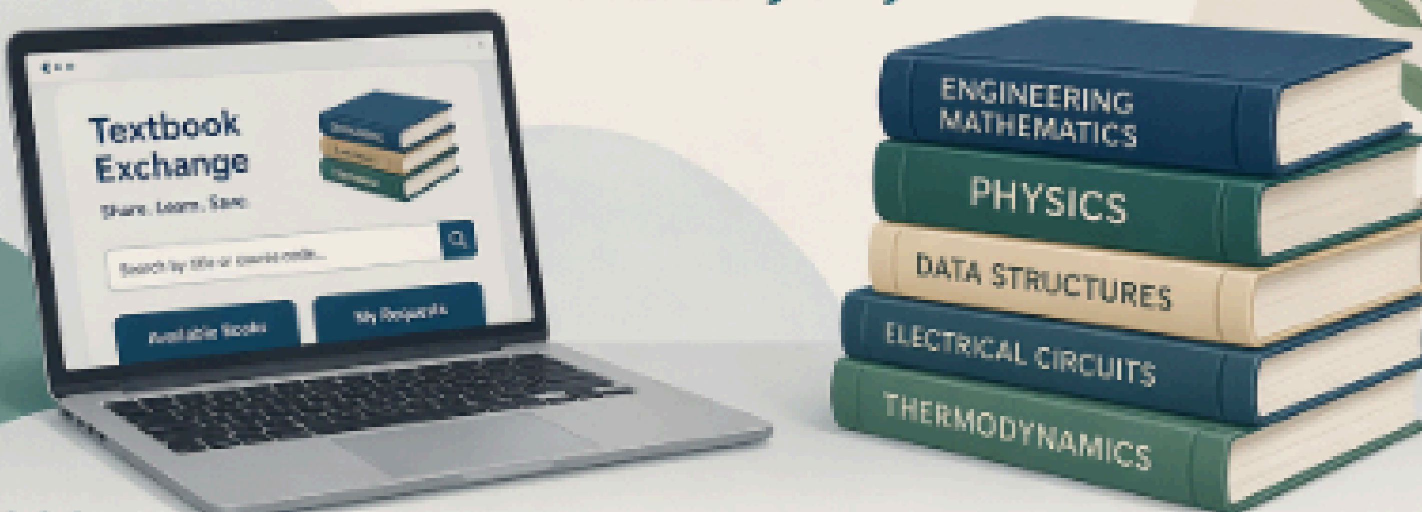
Engineering Club

SOFTWARE PROJECT REPORT

Exchanging University Textbooks Between University Students in an Easy Way



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"Share a book,

1. Introduction

This report presents the detailed requirements, system architecture, user stories, and testing strategy for the University Book Exchange System. The system is designed to help students exchange textbooks efficiently and reduce financial burden.

2. Software Requirements

2.1 Functional Requirements

❖ User Management

1) Registration & Login:

- The system shall allow users to create a new account using an email and password.
- The system shall provide a "Quick Login" feature via email accounts (e.g., Google login) to simplify access.

2) Profile Management:

- The system shall allow users to edit their personal information, such as name, phone number.

3) Account Recovery:

- The system shall provide a password recovery mechanism by sending a verification code to the user's email.

❖ Book Management

1) Add New Books:

- The system shall allow the admin to add new books with details.

2) Update Book Info:

- The system shall allow updating book information or changing their status (e.g., Available, Borrowed).

3) Delete/Archive

- The system shall provide an option to delete or archive books if they are damaged or removed from the library.

❖ Search & Filtering

1) **Advanced Search:**
   - The system shall enable users to search for books by title, or keywords.
2) **Filter:**
   - The system shall allow users to filter search results by academic department (e.g., Computer Engineering, Electrical Engineering...).

❖ Borrowing & Reservation

1) **Borrowing Request:**
   - The system shall allow the user to submit a borrowing request for "Available" books only.
2) **Reservation/Waitlist:**
   - The system shall allow users to reserve a book that is currently borrowed and notify them once it is returned.
3) **Borrowing Limits:**
   - The system shall prevent users from borrowing more than 3 books at a time.

❖ Notifications

1) **Return Reminders:**
   - The system shall send an automatic notification to the user 24 hours before the return deadline.

❖ Reports & Analytics:

- The system shall generate periodic reports showing the most borrowed books in each engineering department to identify student needs.
- The system shall generate a report of books with long waitlists to help the college decide on purchasing additional copies.
- The system shall allow students to view a history of all their previously borrowed books.

2.2 Non-Functional Requirements

❖ Usability

- The system shall have a simple and user-friendly interface.

- ❖ Language Support
  - The system shall support both Arabic and English.
- ❖ Responsiveness
  - The system shall be responsive and work on mobile and desktop devices
- ❖ Performance
  - The system shall load pages quickly within a few seconds.
  - The system shall handle multiple users efficiently.
- ❖ Security
  - The system shall protect user data and login information.
  - The system shall restrict admin features to authorized users only.
- ❖ Reliability
  - The system shall store data reliably using a database.
  - The system shall prevent data loss during operations.
- ❖ Maintainability
  - The system shall be easy to maintain and update.
- ❖ Scalability
  - The system shall be able to handle future growth in users and data.

3. Software Architecture

The **University Book Exchange System** follows a **Three-Tier Architecture**, which separates the system into three main layers: Presentation Layer, Application Layer, and Data Layer. This architectural style improves **maintainability, scalability, and security** by separating concerns and allowing each layer to function independently.

4.1 Architecture Overview

The system is designed so that users interact only with the frontend interface, while all processing and data handling are managed in the backend and database layers.

Flow of interaction:

User → Presentation Layer → Application Layer → Data Layer → Response back to User

4.2 Presentation Layer (Frontend)

The Presentation Layer is responsible for handling all user interactions with the system. It provides a clean and user-friendly interface that allows students and administrators to perform their tasks quickly.

Technologies Used:

- HTML5
- CSS3
- Bootstrap

Responsibilities:

- Displaying available books and search results
- Providing forms for login, registration, and book submission
- Allowing users to reserve books
- Displaying system feedback (success/error messages)
- Ensuring responsive design for mobile and desktop devices

4.3 Application Layer (Backend)

The Application Layer contains the core business logic of the system. It processes user requests, applies system rules, and communicates with the database.

Technology Used:

- PHP

Responsibilities:

- Handling user authentication and session management
- Processing book-related operations (add, edit, delete)
- Managing reservation and approval workflows
- Updating book status (Available, Pending, Borrowed, Damaged)
- Validating user inputs and ensuring data integrity

4.4 Data Layer (Database)

The Data Layer is responsible for storing and managing all system data using a relational database.

Technology Used:

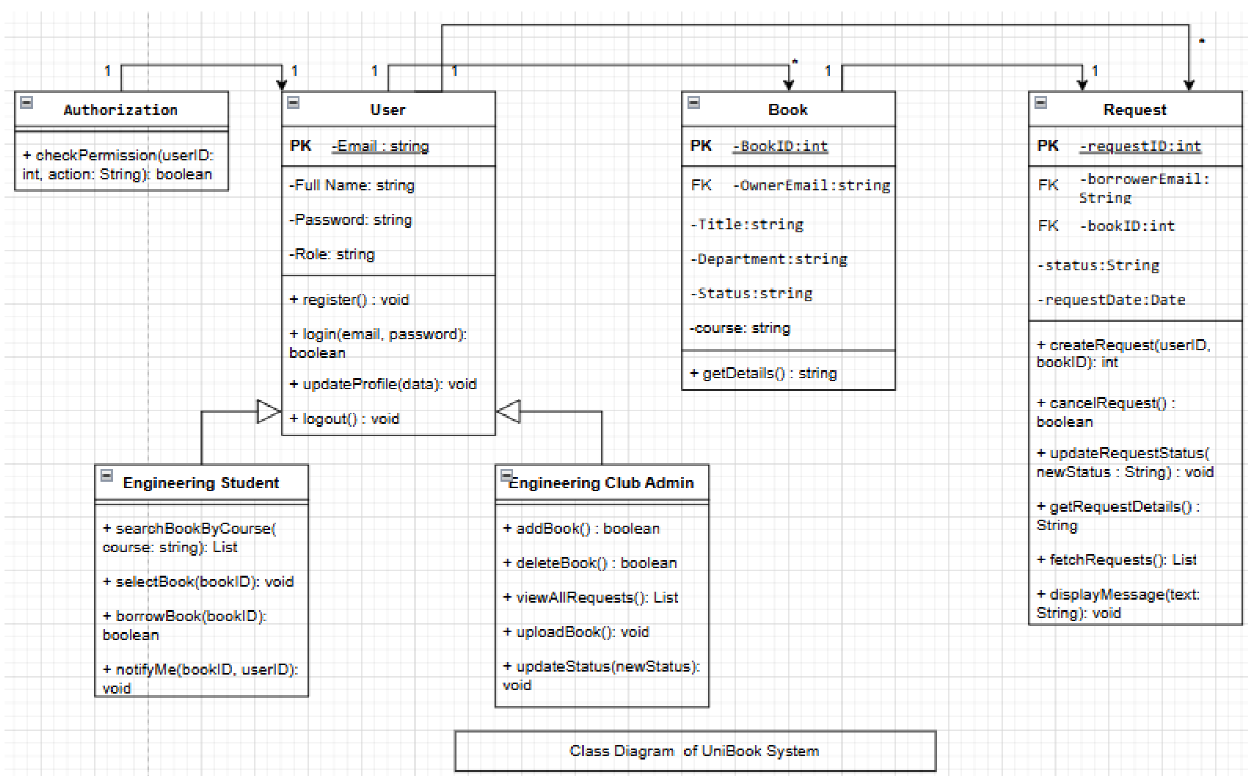
- MySQL
- postgreSQL

Responsibilities:

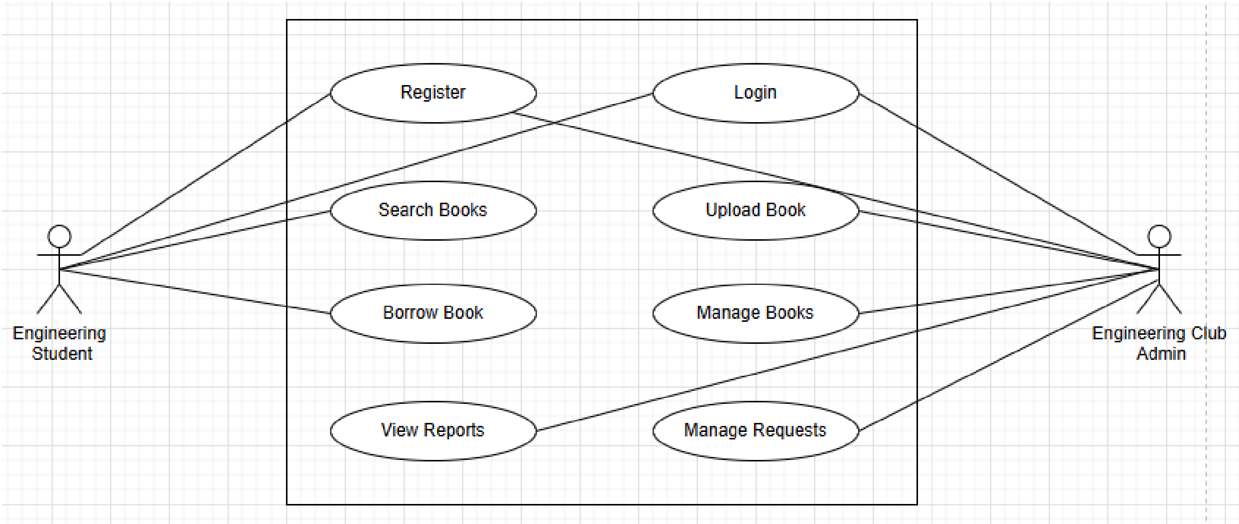
- Storing user information (students, admins)
- Managing book inventory data
- Recording reservation and borrowing transactions
- Ensuring data consistency and integrity
- Supporting efficient querying and data retrieval

5. UML Diagrams

1)UML Class Diagram:

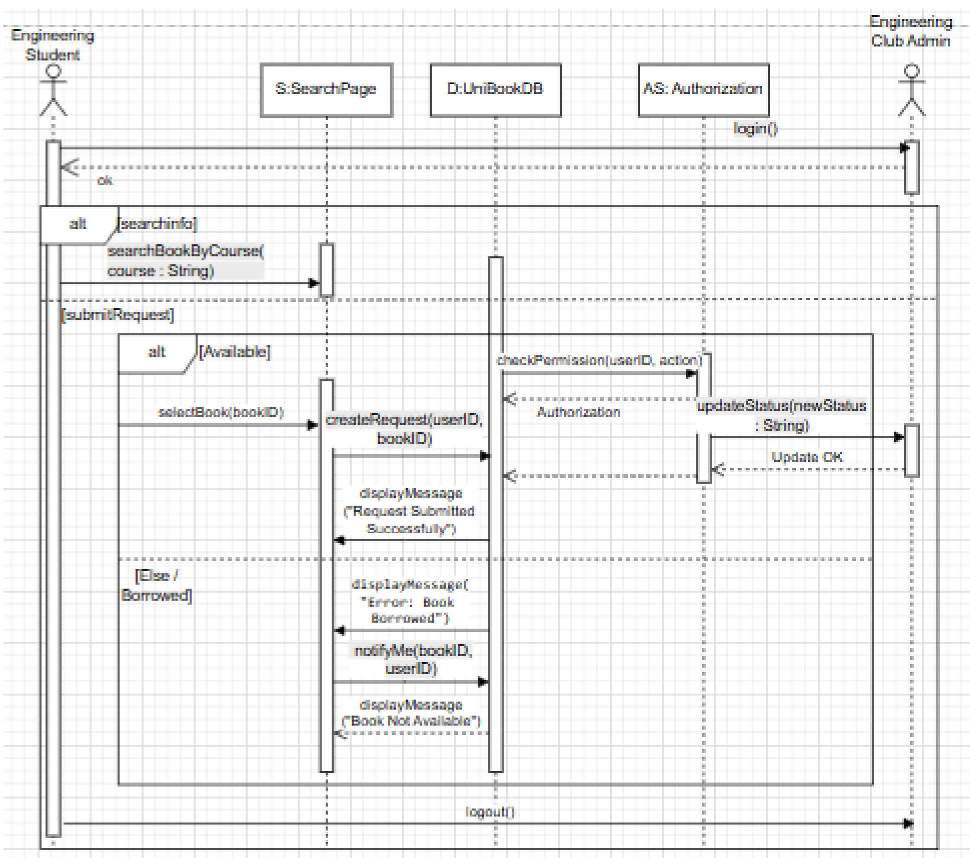


2) Use Case Diagram:



Usecase diagram of UniBook System

3) Sequence Diagram:



Sequence Diagram of UniBook System

6. User Stories

- As a Student Hadeel Irheem , I want to register and log in so that I can access my account and use the system .
- As a Student Reem Nassar , I want to search for books so that I can easily find them by title or course name.
- As a Student Hadeel Irheem, I want to view available books so that I can see which books are available for borrowing.
- As a Student Reem Nassar , I want to suggest books so that I can recommend useful books to be added to the system .
- As a student Hadeel Irheem , I want to reserve a book so that I can secure it .
- As a student Reem Nassar , I want to view the status of my requests, so that I can know whether they are pending, approved, or rejected.
- As a student Hadeel Irheem , I want to receive approval notifications, so that I am informed when my request has been approved.
- As an admin Reem Nassar , I want to log in to the dashboard, so that I can access and manage the system using valid credentials.
- As an admin Hadeel Irheem, I want to view all requests, so that I can see all reservation requests in the system.
- As an admin Reem Nassar, I want to approve or reject requests, so that I can update the request status correctly.
- As an admin Hadeel Irheem, I want to add books, so that the books are successfully stored in the database.
- As an admin Reem Nassar, I want to update book status, so that the status is correctly changed (Available, Borrowed, etc.).
- As an admin Hadeel Irheem, I want to remove damaged books, so that they are deleted from the system.
- As an admin Reem Nassar, I want to view analytics, so that I can see the most requested books.

7.Scenarios

Scenario: Borrowing a Textbook for a Specific Course

- **Objective:** To improve book accessibility and reduce costs for engineering students.
- **Actor:** Hadeel Irheem(Engineering Student).

- **Pre-conditions:** Sarah has logged into the system, either through her existing account or via the "Quick Registration/Sign-up" option.

Steps (Success Scenario):

1. **Search:** Sarah enters the name of her course (e.g., "Digital Logic Design") in the search bar.
2. **Display:** The system filters and displays a list of available textbooks associated with that specific course.
3. **Selection:** Sarah selects the desired textbook and views its details, ensuring the status is "Available".
4. **Request:** She clicks on the "Borrow" button and confirms the borrowing request.
5. **Confirmation:** The system validates the request, updates the book's status in the database to "Borrowed," and sets a return deadline.
6. **Efficiency:** The system automatically notifies the Engineering Club management to facilitate the physical hand-over of the book, enhancing operational efficiency.

Scenario: Book Not Available (Alternative Flow)

- **Objective:** To handle cases where a book is not found and maintain student engagement.
- **Actor:** Reem Nassar (Engineering Student).

Steps:

1. **Search:** Student searches for a book by course name or title.
2. **System Check:** The system finds no books matching the search criteria.
3. **Response:** The system displays a "Not Found" message and offers a "Notify Me" option.
4. **Action:** Student clicks "Notify Me" to be alerted when a copy becomes available.

Scenario: Returning a Borrowed Book

- **Objective**: To update book status and inventory after manual return.
- **Actors**: Student and Engineering Club Admin.

Steps:

1. **Hand-over**: Student brings the physical book back to the Engineering Club.
2. **Identify**: Admin searches for the specific Request ID in the system.
3. **Validation**: Admin clicks "Confirm Return" after inspecting the book condition.

4. **Update**: The system updates the book status to "Available" in the database.

Scenario: Adding a New Book (Admin Flow)

- **Objective**: To expand the database of available textbooks for students.
- **Actor**: Engineering Club Admin.

Steps:

1. **Access**: Admin logs into the management dashboard.
2. **Selection**: Admin selects the "Add New Book" option.
3. **Data Entry**: Admin enters book details including Title, Course, and Department.
4. **Finalization**: The system saves the book and makes it visible for student searches.

Scenario: Handling Overdue Books (System Constraint)

- **Objective**: To maintain management efficiency and ensure timely book recovery.
- **Actor**: The System (Automated).

Steps:

1. **Scan**: The system scans the database for books past their return deadline.
2. **Update**: The system changes the request status to "Overdue" automatically.
3. **Alert**: The system sends an automated email notification to the student.

Scenario: Canceling a Borrow Request

- **Objective**: To allow students to cancel a request before physical pick-up and free up the book for others.
- **Actor**: Student Hadeel Irheem.

Steps:

1. **View Requests**: Students navigate to their "Active Borrowing Requests" section.
2. **Selection**: Students select the specific pending request they wish to cancel.
3. **Cancellation**: Student clicks the "Cancel Request" button and confirms the action.
4. **Update**: The system deletes the request and updates the book status back to "Available" in the database.

Scenario: Updating User Profile

- **Objective**: To enable users to maintain accurate and up-to-date personal information.
- **Actor**: Student or Admin.

Steps:

1. **Access**: User navigates to the "Profile Settings" or "Account Management" page.
2. **Edit**: User modifies their personal details (e.g., Full Name, Password, or Contact Info).
3. **Submission**: User clicks the "Save Changes" button.
4. **Validation**: The system validates the input data and updates the user's record in the database.

Scenario: Request Rejection by Admin

- **Objective**: To provide the administration with control over borrowing permissions based on student records.
- **Actor**: Engineering Club Admin.

Steps:

1. **Review**: Admin reviews a new incoming borrow request from the dashboard.
2. **Decision**: Admin identifies an issue (e.g., student has too many overdue books) and clicks "Reject Request."
3. **Reasoning**: Admin enters a brief reason for the rejection (optional).
4. **Notification**: The system notifies the student of the rejection and keeps the book status as "Available" for other users.

8. Software Testing

ID	Test Case	Steps	Expected Result
TC1	Login	Enter credentials	Login success
TC2	Add Book	Fill form	Book added
TC3	Reserve Book	Click reserve	Status = Pending

TC4	Approve Request	Admin approves	Status = Borrowed

9. Conclusion

The system improves book accessibility, reduces costs for students, and enhances management efficiency for the Engineering Club.To ensure professional execution, the team utilized **Jira** for agile task tracking, **GitHub** for version control, and **Draw.io** for designing the UML architecture, ensuring a scalable and reliable platform for the university community.